Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	KAI
Project / Solution Name	One Stop Smart Procurement
Name (Leader)	Woon Kai Xiang
Email & Phone (Leader)	woonkx911@gmail.com 016-7230900
University Affiliation(s)	Sunway University
Live Prototype URL	One Stop Smart Procurement
Code Repository URL (public / access granted to judges)	GetGood-source/SDG2026-KAI

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Our application connects its components through the following workflow:

1. **User interface:** Employees and managers access the React and TypeScript web app. An employee enters the required part, vehicle model, quantity and deadline.
2. **Backend:** The interface sends the request to the Express backend through a REST API. The backend validates the details and checks inventory and supplier records stored in PostgreSQL.
3. **Supplier discovery:** If no suitable recorded option is available, the backend uses SerpApi to retrieve Google Search and Maps supplier leads. Staff must confirm the supplier's actual quote, stock and part compatibility.
4. **Comparison and AI explanation:** Rules-based logic ranks recorded quotations. GPT-5.4 mini, accessed through Replit AI Integrations, can explain the comparison using the supplied evidence. It does not invent prices or approve purchases.
5. **Approval and storage:** A manager approves or rejects the purchase. The backend saves the decision, internal purchase status and audit information in PostgreSQL through Drizzle ORM, then returns the updated information to the interface.
6. **Cloud hosting:** Replit hosts the web interface and backend so users can access the published prototype through a browser.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Drizzle ORM 0.45.4	Reads and writes PostgreSQL data	Apache-2.0
TypeScript 5.9.3	Programming language used for frontend and backend development	Apache-2.0
React 19.1.10	Builds the interactive user interface	MIT

Name & Version	Purpose in Project	License
Vite 7.3.7	Development server and frontend build tool	MIT
Express 5.3.0	Handles backend API requests and procurement actions	MIT
Tailwind CSS 4.3.3	Styles the app's layout, colours and responsive interface	MIT
Radix UI Dialog 1.2.0	Provides accessible dialog components	MIT
TanStack React Query 5.104.1	Fetches, caches and refreshes backend data	MIT
node-postgres (pg) 8.23.1	Connects the backend to PostgreSQL	MIT
Zod 3.25.76	Validates input and API response data	MIT
Wouter 3.13.0	Handles navigation between pages	Unlicense
Lucide React 0.545.0	Provides interface icons	ISC
OpenAI SDK 6.49.0	Calls the AI service for quotation explanations	Apache-2.0
Clerk React SDK 6.18.0 and Express SDK 2.1.81	Supports account-enabled authentication; the current demo bypasses sign-in	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenAI-compatible inference API through **Replit AI Integrations**	Generates explanations of recorded quotations and rules-based recommendations	Usage-based; deducted from Replit credits. Sponsor-provided status unverified
SerpApi — Google Search and Google Maps results	Discovers potential suppliers, business details, ratings, source links and available advertised-price evidence	Own API key; free/paid account tier unverified
Clerk API and hosted SDK services	Supports the optional account-enabled mode. Included SDKs may initialize, but the public demo does not require account verification	Replit-managed tenant; service tier unverified

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Publishing	Hosts the frontend's production files and runs the Express backend	**Autoscale deployment; successful build confirmed.** Subscription tier and region unverified
PostgreSQL database connected to the Replit project	Persists requests, inventory, suppliers, quotations, approvals, settings and audit records	Project-connected database; hosting plan and region unverified

Service & Provider	Purpose in Project	Plan / Region
Replit AI Integrations	Provides server-side access to the AI model	Usage billed through Replit credits; processing region unverified
Clerk — Replit-managed tenant	Supports secure account-enabled operation; not required for demo role entry	Tenant plan and region unverified

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
GPT-5.4 mini (gpt-5.4-mini)	OpenAI through Replit AI Integrations	Pre-trained hosted model used to explain recorded quotation comparisons. Not fine-tuned or self-trained	Proprietary hosted model; governed by applicable OpenAI and Replit service terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Sample procurement records: suppliers, inventory, quotations and requests	Project-generated seed data	Synthetic	Project-created; not an externally licensed dataset	**N**, intended as fictional demonstration data

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Live supplier-discovery results	Google Search/Maps results retrieved through SerpApi	Publicly accessible third-party data	Governed by provider and source-data terms; not an openly licensed dataset	Y, potentially—public contact details may identify sole traders
User-entered requests, confirmed quotes, supplier records and audit history	Entered directly into the application	Self-collected / private business data	Internal project/business data; no public license assigned	**Y, potentially**—names, contacts or free-text fields may contain personal information
Account and team records, when account-enabled mode is used	Clerk and manager-entered membership information	Private / self-collected	Governed by application policies and provider terms	**Y**—names, email addresses and user identifiers; demo identities are simulated

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Replit Agent — cloud service, version not specified	AI-assisted code generation, interface design, backend development, debugging, implementation refinements, technical explanations and drafting project documentation	**Major**

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team reviewed AI-generated code, interface designs and written content against the project requirements rather than accepting them unchanged. We checked the prototype through type checks, production builds, API responses and visual reviews, then refined issues identified through feedback.

For the app's AI explanations, we restricted the model to recorded quotations and checked that supplier rankings remain rules-based. Google results are labelled as unconfirmed leads; staff must confirm quotes, and a manager must approve purchases.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	**Replit Agent — AI-assisted development tool**	Helped generate and refine frontend and backend code, design the interface, debug issues and prepare documentation.
Replit	**Cloud development workspace and Publishing platform**	Used to develop, run, build and publish the web interface and backend API.
Replit	**Replit AI Integrations — model API access**	Connects the backend to GPT-5.4 mini to explain recorded quotation comparisons. Include this as sponsor-provided if it was covered by your SDC access.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was central to developing and demonstrating our prototype. Replit Agent supported implementation and refinement, while its cloud platform allowed us to run and publish the application. However, these resources are technically replaceable: the React frontend, Express backend and PostgreSQL database could run on another platform, and a compatible AI API could provide quotation explanations. Replacing Replit would require additional setup, deployment configuration and testing

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The demo stores part requests, inventory, supplier contacts, quotations, approvals and audit entries in PostgreSQL. User-entered names or descriptions may contain personal data. No email login is required, although the included Clerk SDK may contact Clerk during initialization.

SerpApi receives the requested part/model and shop location. Replit AI Integrations/OpenAI receives part details, quantity, deadline and recorded quotations for AI explanations. API credentials remain server-side in Replit Secrets, and the published site uses HTTPS.

This is an open, shared demo: anyone can choose Manager and access shared records. Users should not enter real customer, personal or confidential information.

There is no automatic retention deadline. Application records remain until manually deleted. Clearing demo history removes requests, quotations and audit entries but retains suppliers, inventory and settings. Third-party retention follows each provider's terms.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The AI may produce incomplete or inaccurate explanations. To reduce this risk, it is instructed to use only recorded quotations; supplier ranking is rules-based, and AI cannot select or approve a purchase.

Live Google results via SerpApi may be incomplete, outdated or irrelevant. A listing does not prove exact-part fitment, stock, price or delivery. Search may also be unavailable when the provider reaches its quota or has an outage. The app labels results as unconfirmed leads, retains source links, requires staff to confirm quotes, and requires manager approval. Manual quote entry remains available.

The prototype depends on external AI and search providers, so their outages, rate limits or service changes can affect those features. The public demo has simulated roles: anyone can choose Manager. It should contain no real customer or confidential business data.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] ***KAI***

[Date] 25/10/2026